

MAT 524: Functions of a Real Variable II

Homework VII

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[[Problem Three]]:

Problem Statement

Using Bessel's inequality, prove that for every measurable subset $A \subseteq [0, 1]$,

$$\lim_{n \rightarrow \infty} \int_A \cos(2\pi nx) dx = \lim_{n \rightarrow \infty} \int_A \sin(2\pi nx) dx = 0.$$

Solution

We begin by noting that the function $y = \frac{\chi_A}{\sqrt{2}}$ is in $L^2([0, 1])$ and the set

$$\bigcup_{n=1}^{\infty} \left\{ \sqrt{2} \cos(2\pi nx) \right\}$$

is orthonormal in $L^2([0, 1])$. For $n \in \mathbb{N}$, the Fourier coefficients of y are

$$\hat{y}(n) = \left\langle y, \sqrt{2} \cos(2\pi nx) \right\rangle \tag{1}$$

$$= \int_0^1 \frac{\chi_A}{\sqrt{2}} \sqrt{2} \cos(2\pi nx) dx \tag{2}$$

$$= \int_A \cos(2\pi nx) dx. \tag{3}$$

By Bessel's inequality,

$$\sum_{n=1}^{\infty} \left| \int_A \cos(2\pi nx) dx \right|^2 \leq \|y\|^2.$$

But

$$\|y\|^2 = \left(\left[\int_0^1 \left| \frac{\chi_A}{\sqrt{2}} \right|^2 \right]^{1/2} \right)^2 \tag{4}$$

$$= \int_A \frac{1}{\sqrt{2}} \tag{5}$$

$$= \frac{m(A)}{\sqrt{2}} \tag{6}$$

$$< \frac{1}{\sqrt{2}}. \tag{7}$$

Thus, the sum $\sum_{n=1}^{\infty} \left| \int_A \cos(2\pi nx) \right|^2$ converges, so we must have

$$\lim_{n \rightarrow \infty} \left| \int_A \cos(2\pi nx) \right|^2 = 0.$$

Since the squared absolute value function is continuous, we can move the limit inside, obtaining

$$\left| \lim_{n \rightarrow \infty} \int_A \cos(2\pi nx) \right|^2 = 0,$$

i.e.,

$$\lim_{n \rightarrow \infty} \int_A \cos(2\pi nx) = 0.$$

Now, since

$$\bigcup_{n=1}^{\infty} \left\{ \sqrt{2} \sin(2\pi nx) \right\}$$

is also orthonormal in $L^2([0, 1])$, we can apply the same argument as above to conclude that

$$\lim_{n \rightarrow \infty} \int_A \sin(2\pi nx) = 0.$$

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